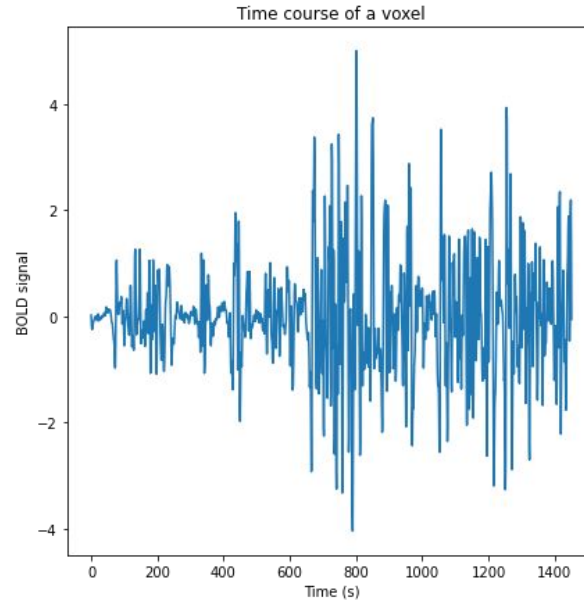
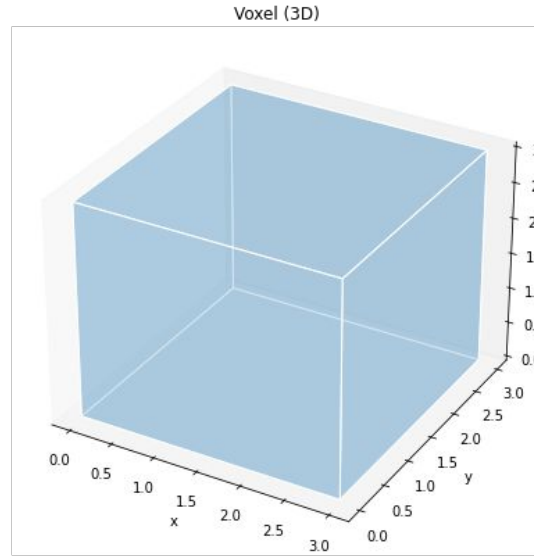
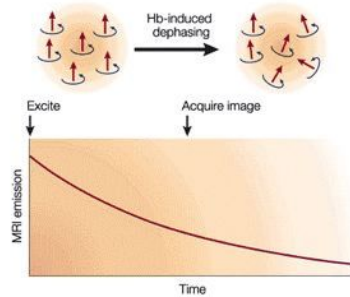
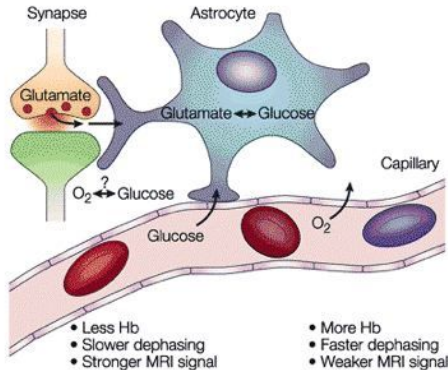


Predicting ADHD from Resting-State fMRI: *“Connectivity Patterns Using Machine Learning”*



Introduction: Resting-State fMRI

BOLD (blood oxygen-level dependent) signal

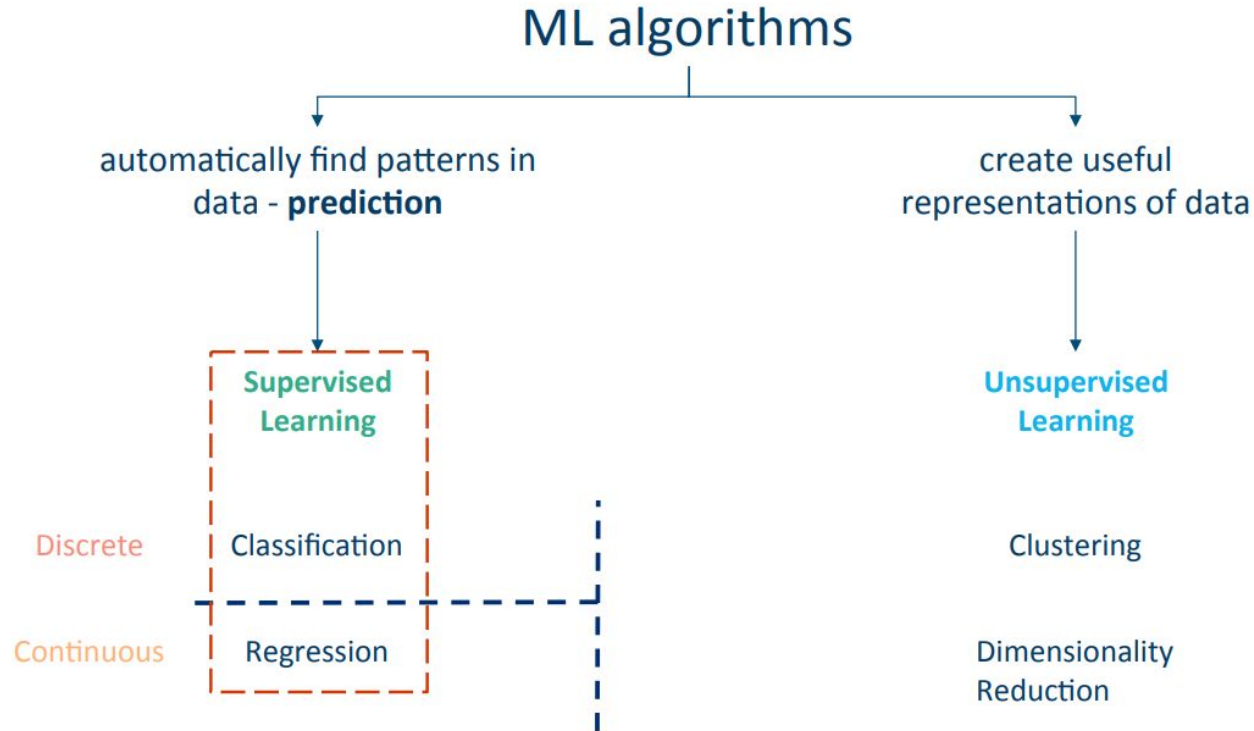


Non-invasive method to explore intrinsic brain networks in neurological and neurodevelopmental disorders.

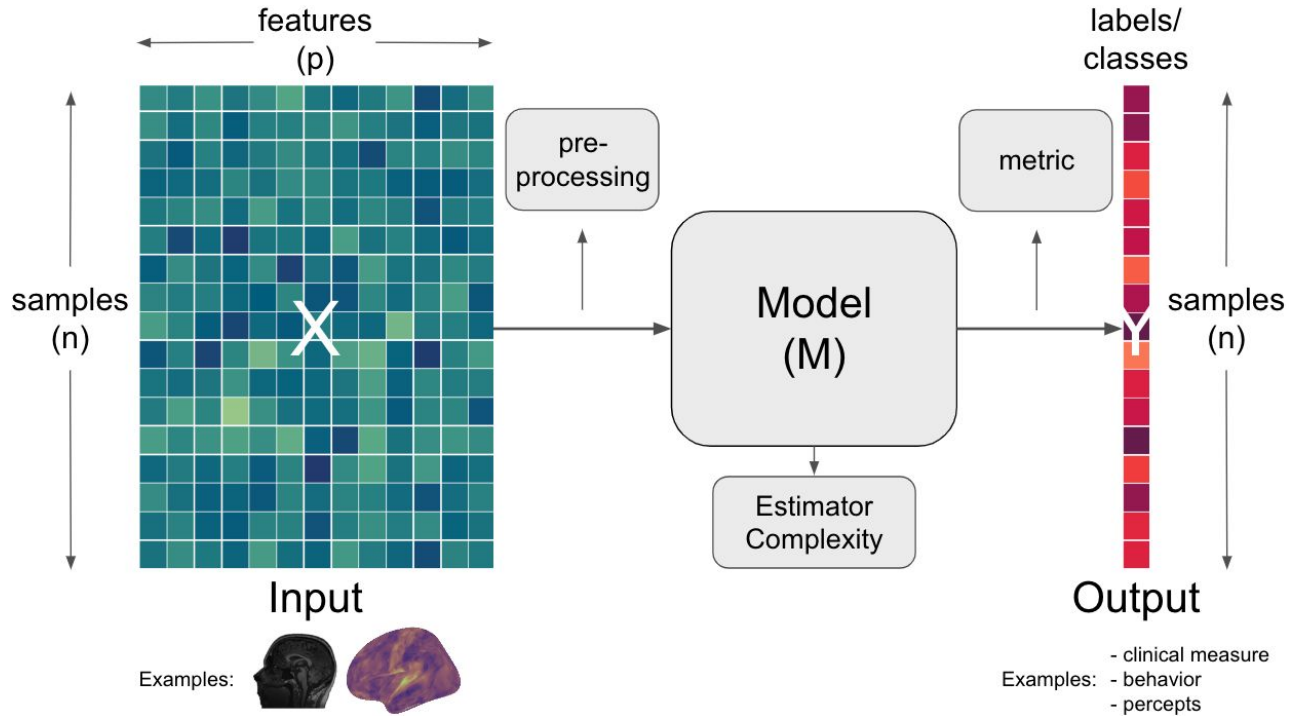
Echo-planar imaging (EPI) reduces imaging time compared to conventional MRI.

Participants rest with eyes closed during rs-fMRI to capture brain activity without task interference.

Introduction: Machine Learning

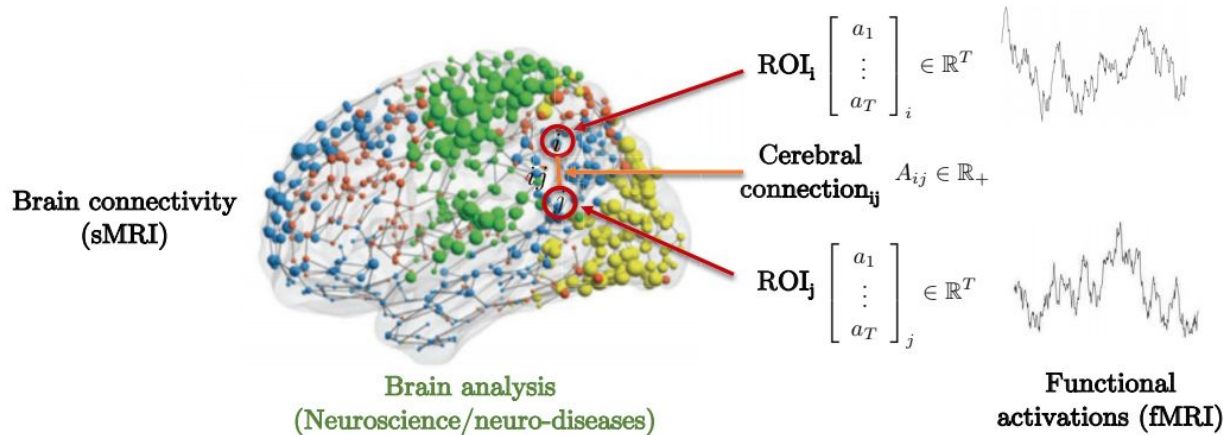


Introduction: Machine Learning



Hypothesis

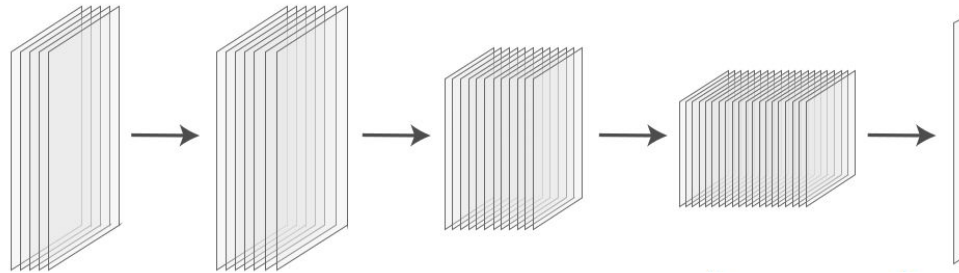
Individuals with a specific disorder would exhibit distinct connectivity patterns compared to healthy controls. Using machine learning, this study aims to predict and analyze these patterns, potentially offering data driven improved diagnostic capabilities.



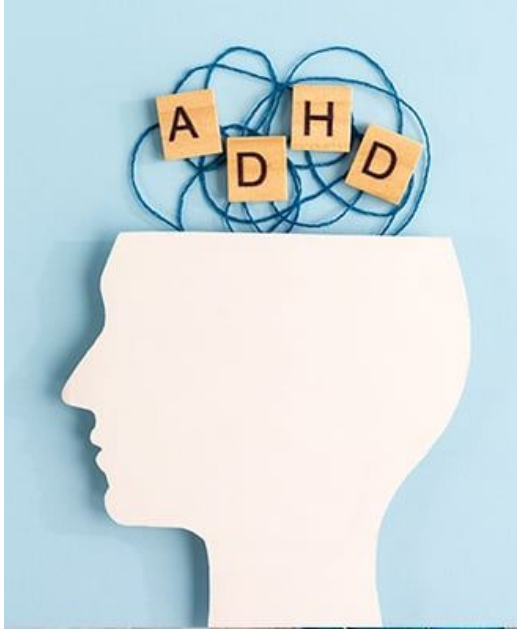
Objective

The aim of the study is to develop a reliable Machine Learning code architecture for classifying psychopathologies using fMRI data, with a specific emphasis on analyzing deep functional brain connectivity in subjects with ADHD.

model



*Methods: **Subjects***



Dataset: ADHD-200 dataset.

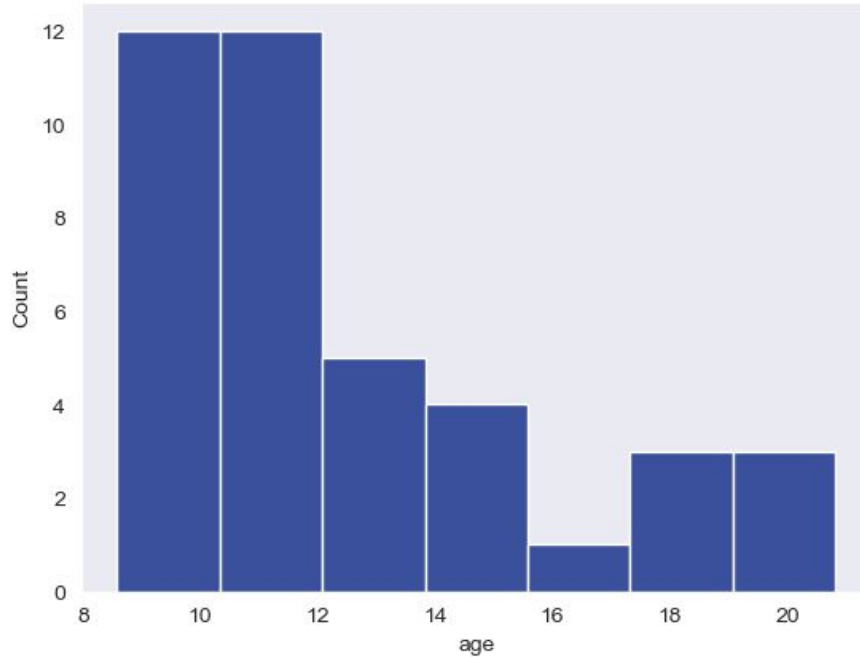
Sample Size: 40 participants (20 diagnosed with ADHD and 20 controls).

Age Distribution: Mean age 12.50 years (range 8.58 to 20.82 years).

Gender Distribution: 35 males and 5 females.

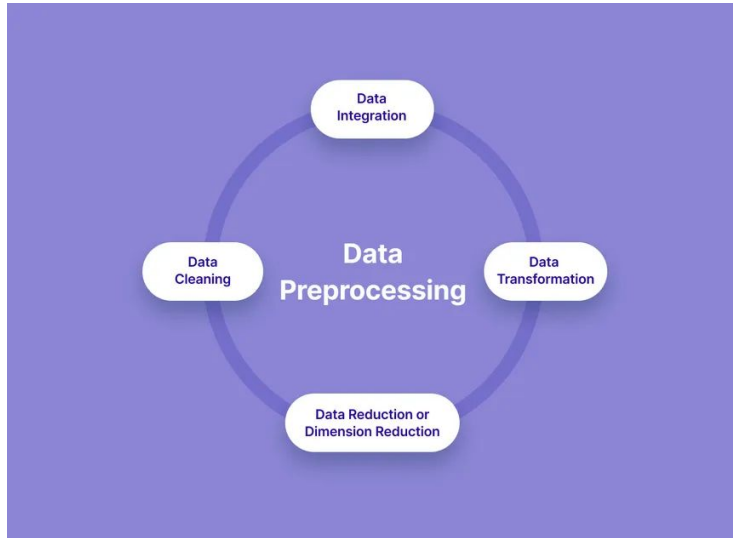
Imaging Technique: Standard echo-planar imaging sequences at various sites.

Fig 1: Age Distribution in the ADHD Dataset



Distribution of ages in the ADHD dataset. The y-axis represents the frequency or density of participants, while the x-axis shows the age range.

Methods: **ADHD-200 dataset**



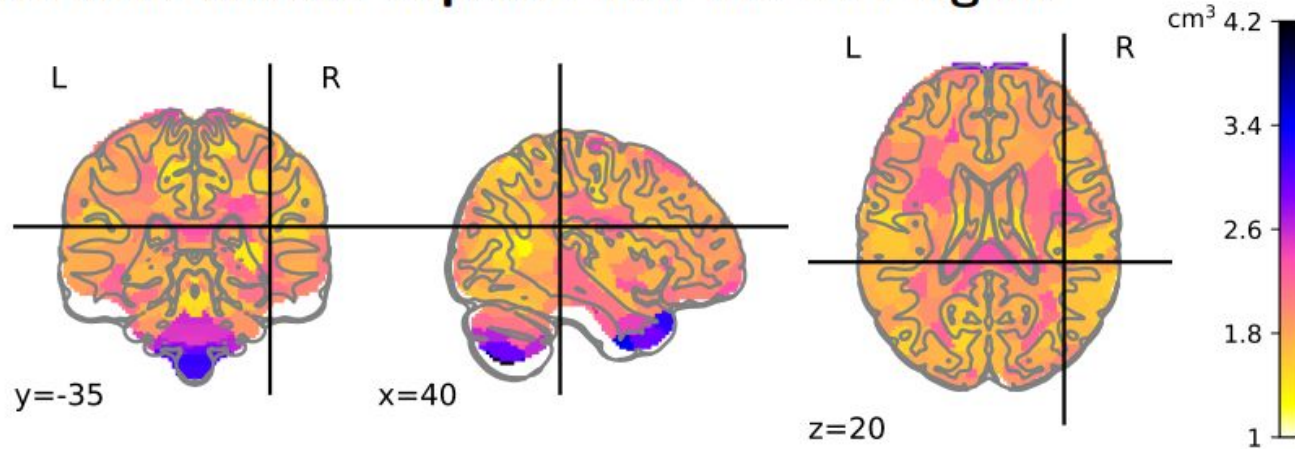
ADHD-200 dataset formatted according to Brain Imaging Data Structure (BIDS).

Preprocessing includes denoising, registration, extraction of regional time series, and generation of intrinsic connectivity networks (ICNs).

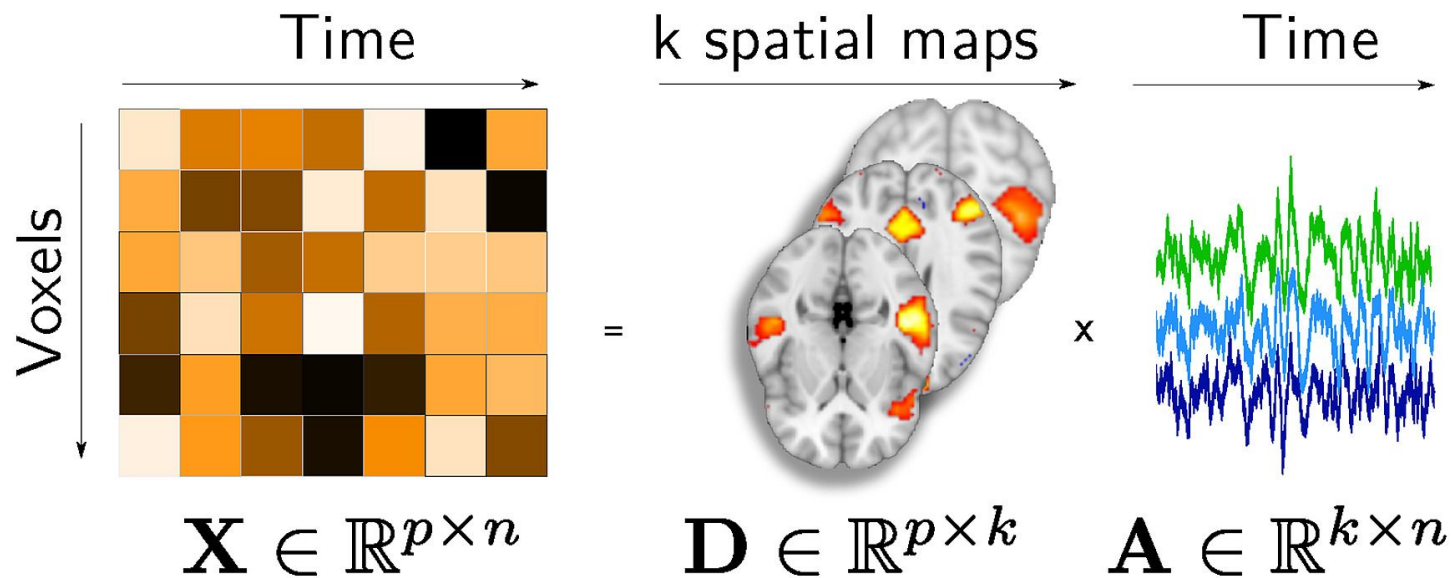
Further analysis includes calculating fractional amplitude of low-frequency fluctuation (fALFF).

Methods: **Echo-planar imaging (EPI)**

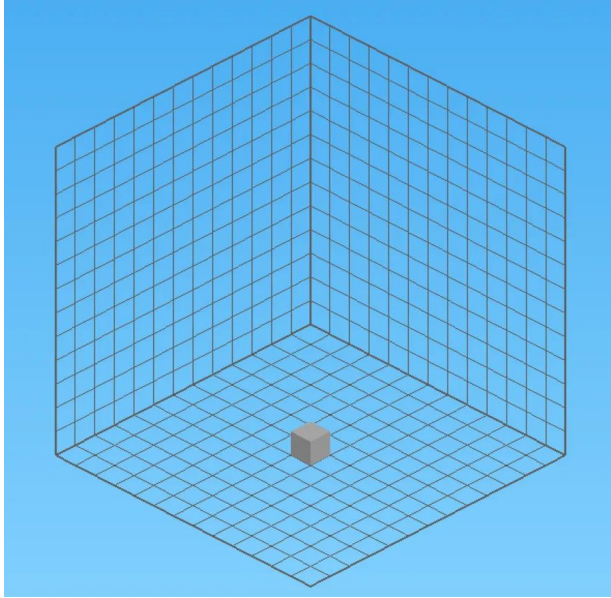
DiFuMo atlases capture well the EPI signal



Methods: DiFuMo atlas (64 ROIs).



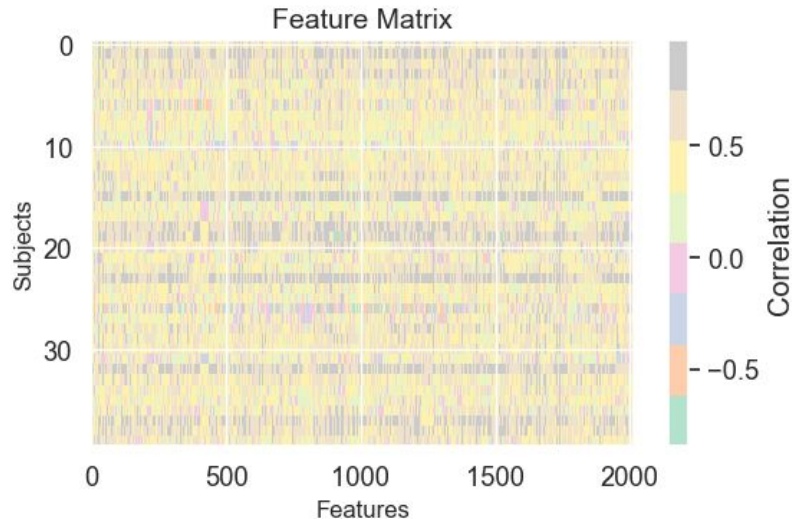
Methods: **Functional Connectivity Analysis**



Connectivity Calculation: Pearson's correlation coefficients between each pair of ROIs, creating a 64x64 connectivity matrix.

Feature Matrix: Shape of (40 subjects, 2016 features), each feature representing the correlation between a specific pair of brain regions.

Fig 2: Feature Matrix Correlations These features are derived from correlation matrices that were transformed into one-dimensional arrays.



X-axis (Columns): This axis would represent the 2016 features. Each feature corresponds to the strength of a connection between a specific pair of brain regions in a participant's data.

Y-axis (Rows): This axis would represent the 40 subjects included in the ADHD dataset. Each row would contain the set of features (connection strengths) for a single participant.

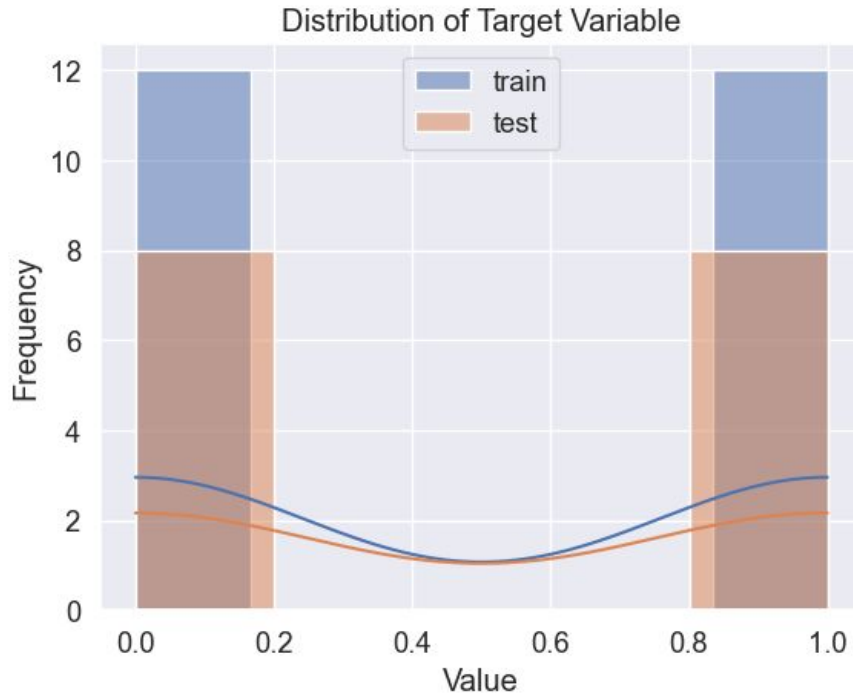
Methods: **Decoding Preprocessing**



Training the Model

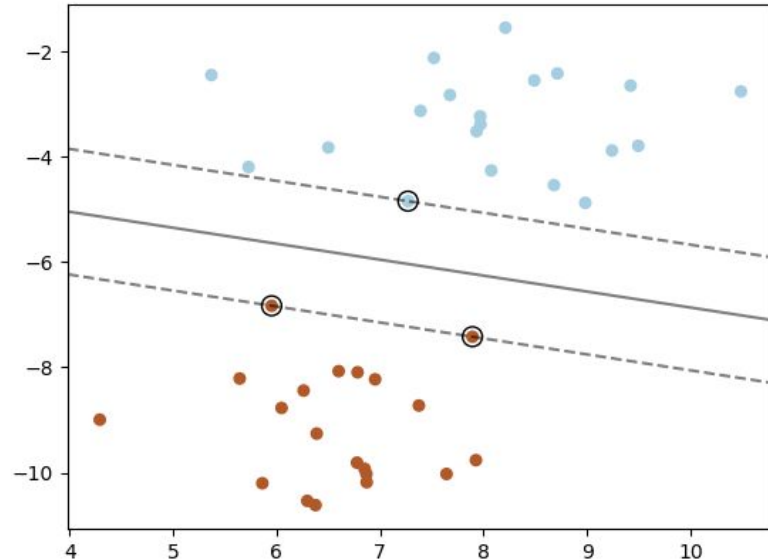
- **Normalization:** Z-score standardization.
- **Data Split:** 60% training set and 40% testing set.
- **Class Balance:** Ensured balanced distribution of ADHD and control across sets.

Fig 3: Training-Testing Split



This figure shows how the ADHD dataset was divided for machine learning. 24 participants were used for the training set, and 16 participants were used for the testing set. The training set helps the model learn patterns, while the testing set evaluates the model's generalizability on unseen data.

Methods: *Decoding Processing*



Support Vector Machine aims at finding an optimal hyperplane to separate two classes in high-dimensional space, while maximizing the margin.

Output: Binary classification (0 for control, 1 for ADHD).

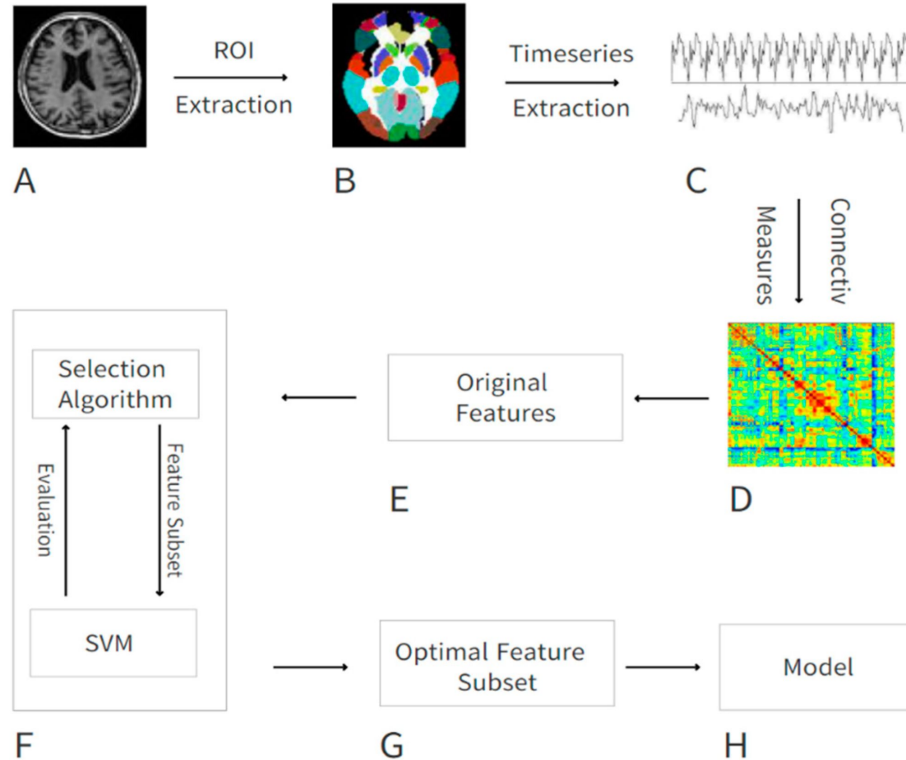
Methods: *Decoding Processing*

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
First Iteration	Test				
Second Iteration		Test			
Third Iteration			Test		
Fourth Iteration				Test	
Fifth Iteration					Test

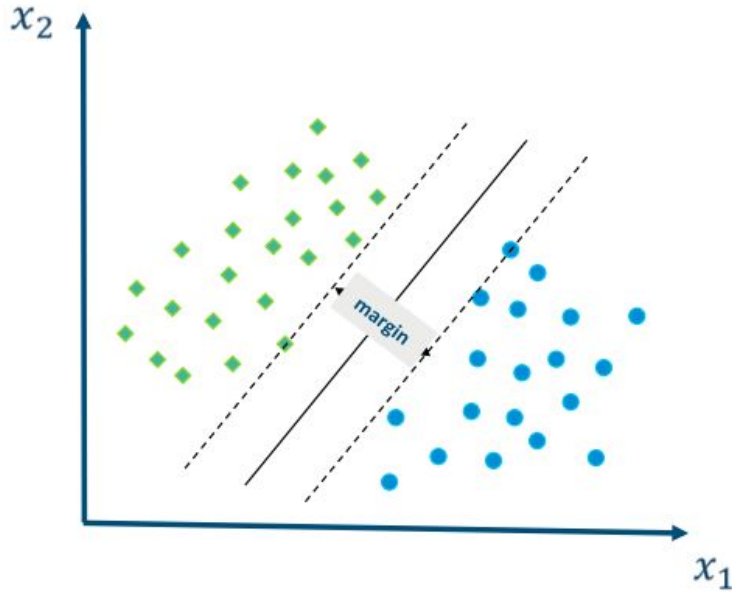
Building and Tuning the SVM Model

- **Hyperparameter Tuning:** Grid search cross-validation to find optimal hyperparameters.
- **Validation:** Cross-validation ensures robust evaluation and prevents overfitting.

Methods: *Decoding Processing Summary*



Results: *Linear Kernel*



Feature Importance: Analysis revealed specific brain connectivity patterns significant to the model's predictions.

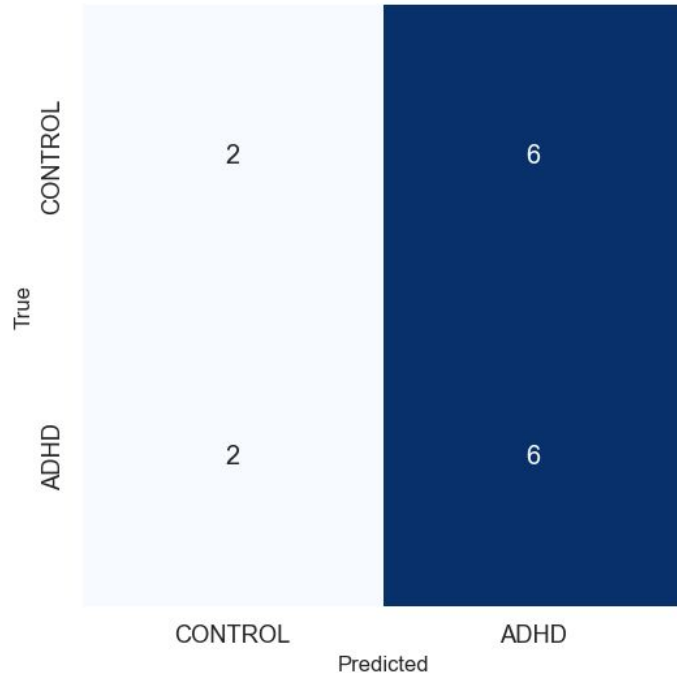
Results: *Linear Kernel*

Classification Report:

	precision	recall	f1-score	support
0	0.50	0.25	0.33	8
1	0.50	0.75	0.60	8
accuracy			0.50	16
macro avg	0.50	0.50	0.47	16
weighted avg	0.50	0.50	0.47	16

SVM Performance: Higher recall for ADHD (0.75) compared to control (0.25).

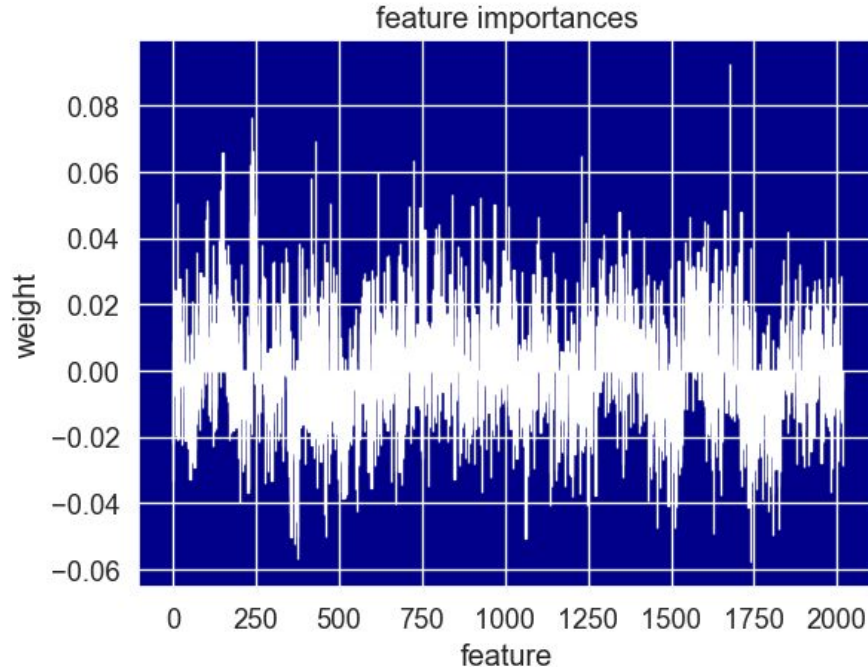
Fig 4: Model Performance on ADHD Classification



The confusion matrix shows how well the machine learning model performed in classifying ADHD.

- The model identified more actual ADHD cases correctly (recall of 75% for ADHD).
- However, the model's precision was similar for both control and ADHD (precision of 50% for both), meaning half of the classifications for each class were correct.

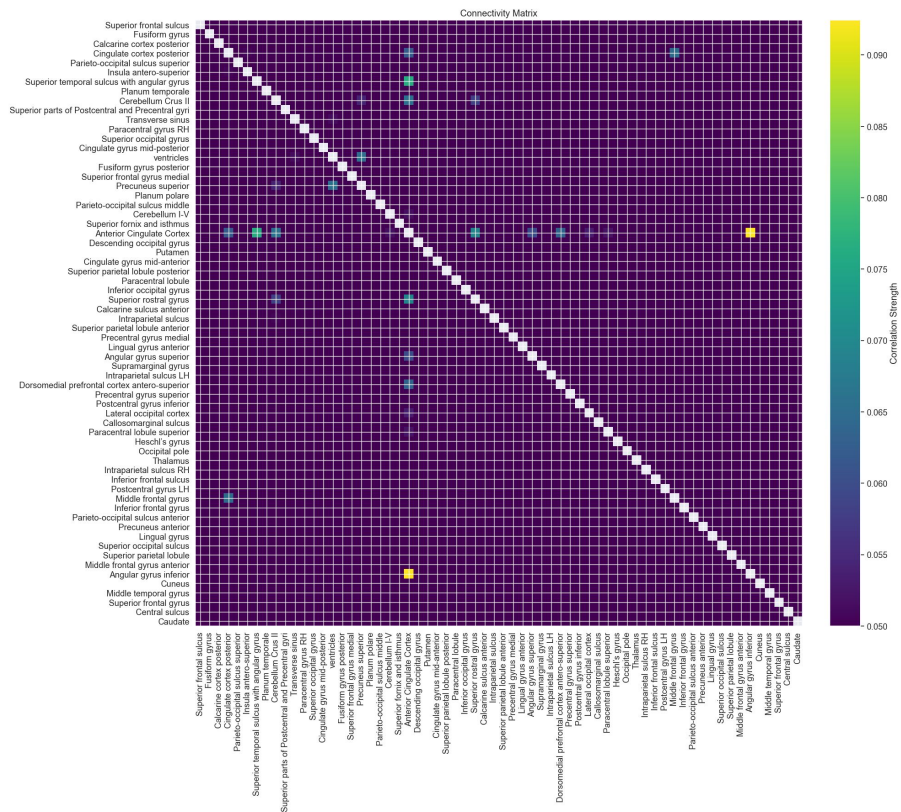
Fig 5: Extracting Insights from Coefficients



The model utilizes coefficients (weights) to determine the significance of each feature (connection between brain regions) for ADHD classification.

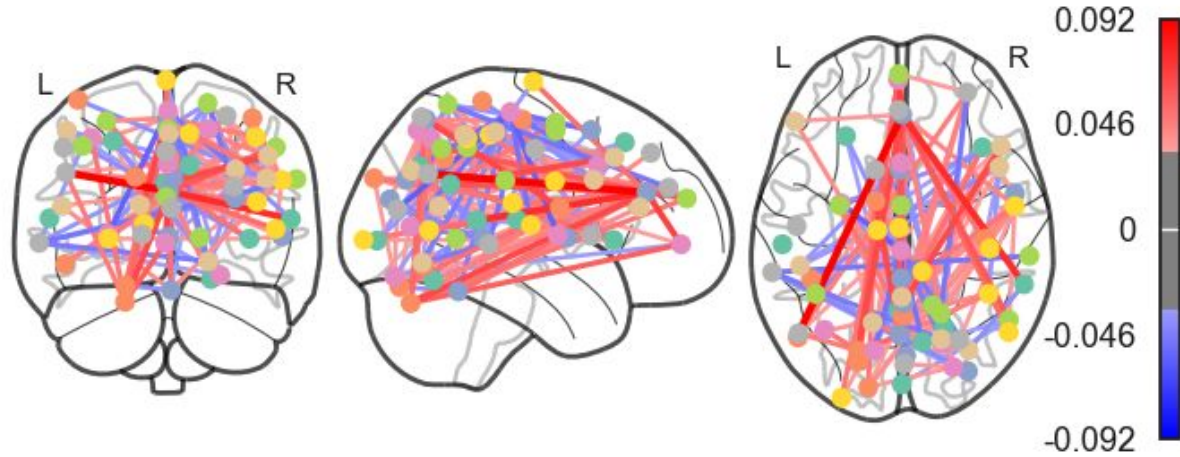
- X-axis: This axis represents the various brain regions analyzed by the model (features).
- Y-axis: This axis represents the weight (coefficient) assigned to each brain region by the SVM model. The height of each bar corresponds to the weight/importance of that specific brain region connection in predicting ADHD. Higher bars indicate a greater influence on the model's predictions.

Fig 6: Regions of Interest Coefficients Matrix



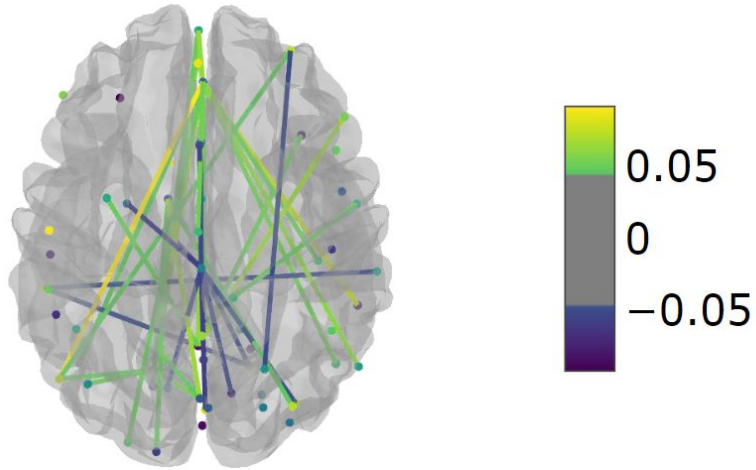
This image shows brain connections that are most important for the SVM model to classify ADHD. It uses a brain atlas (Difumo) that divides the brain into 64 regions. Each square in the image represents a connection between two of these regions. The brighter the square, the stronger the connection's influence on the model's ADHD classification.

Fig 7: Brain Connectivity Network in ADHD



This image shows the strongest connections between brain regions in the ADHD group. Nodes represent brain regions, and thicker lines indicate stronger connections between them.

Results: Connections: Strongest connection identified was between the Anterior Cingulate Cortex (ACC) and the Inferior Angular Gyrus (AGI).

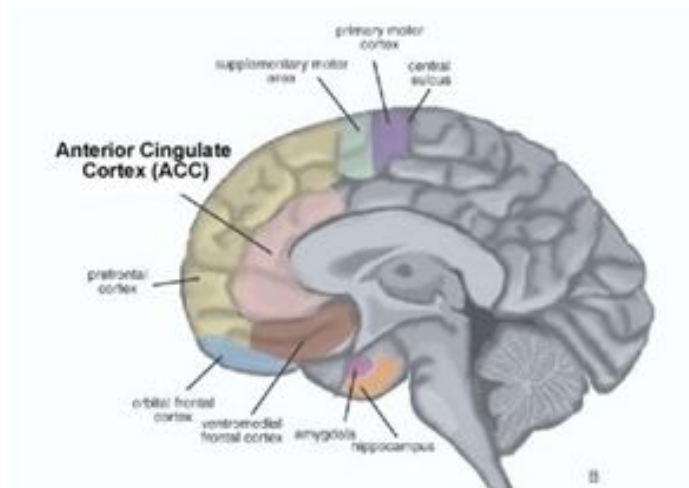


Error detection and conflict monitoring: Together, they may detect and resolve conflicts.

Social cognition: Both could be involved in understanding others' mental states, empathy, and theory of mind.

Cognitive control: They jointly support attention, working memory, and decision-making.

Results: **Connections:** Anterior Cingulate Cortex (ACC)



Error detection and conflict monitoring: Detects errors and conflicts, playing a critical role in monitoring and resolving conflicts.

Emotion regulation: Processes emotional stimuli, emotional arousal, and emotional memory.

Social cognition: Involved in theory of mind, empathy, and understanding others' mental states.

Cognitive control: Manages attention, working memory, and decision-making.

Results: Connections: Inferior Angular Gyrus (AGI)

A lesion in the angular gyrus can lead to:

- **Gerstmann Syndrome:** Characterized by difficulties with arithmetic, writing (agraphia), distinguishing left from right, and finger agnosia.
- **Behavioral Changes:** Reduced activity in this region is observed in dyslexic patients, who face challenges in spelling, reading, and writing.

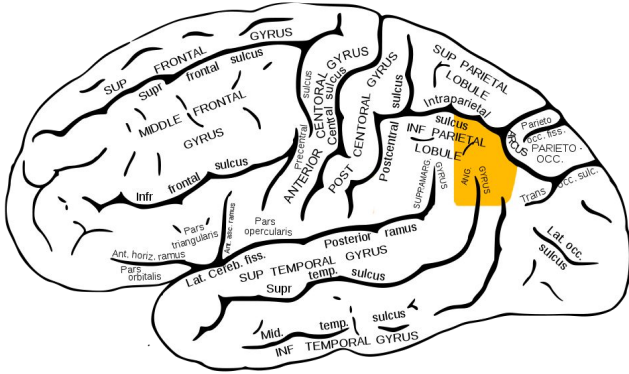
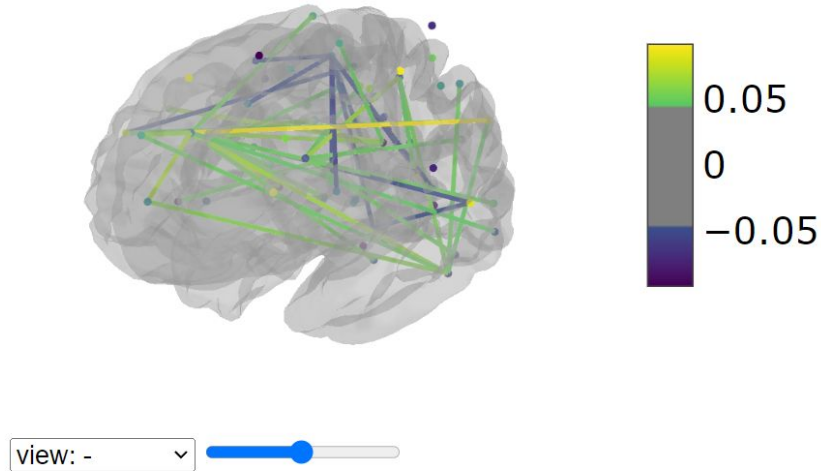


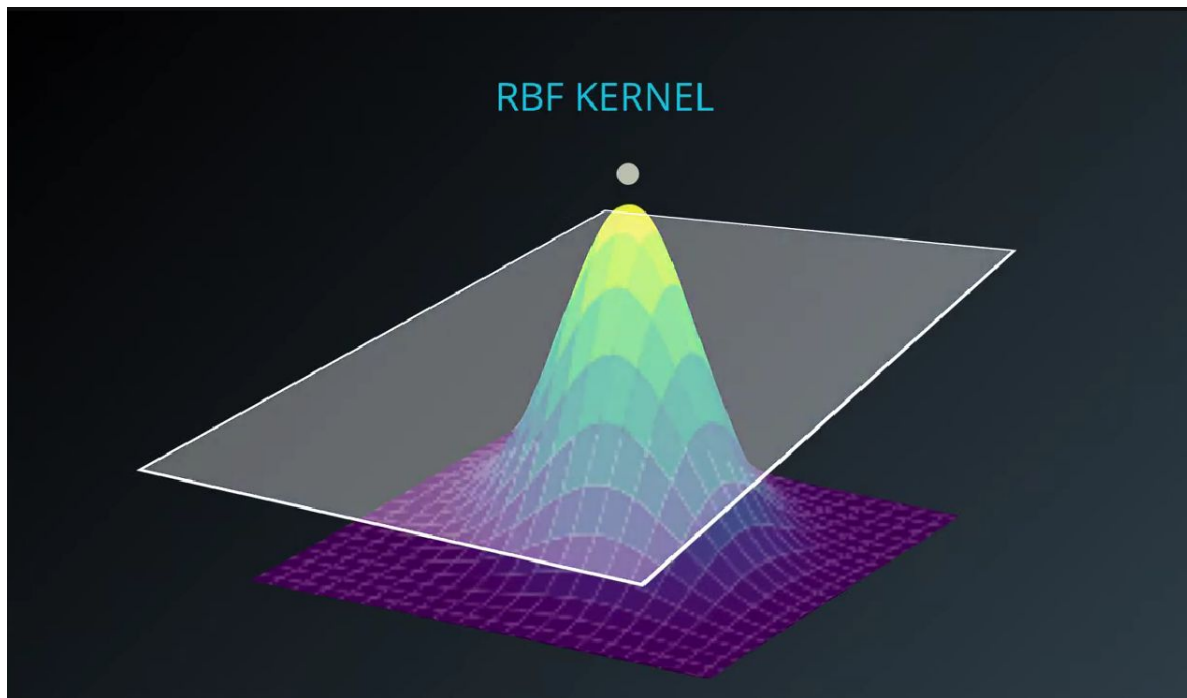
Fig 8: Specific Brain Connectivity in ADHD



Focus on a Specific Connection. Highlights a specific connection identified in this high-threshold connectome:

- **Connection:** Anterior Cingulate Cortex (ACC) to Inferior Angular Gyrus (AGI) (Yellow)
- **Strength:** 0.09245993901160833 (relatively high compared to other connections highlighted in the figure)

*Results: **Non-Linear Kernel***



Results: *Non-Linear Kernel*

Accuracy: 0.6875

Classification Report:

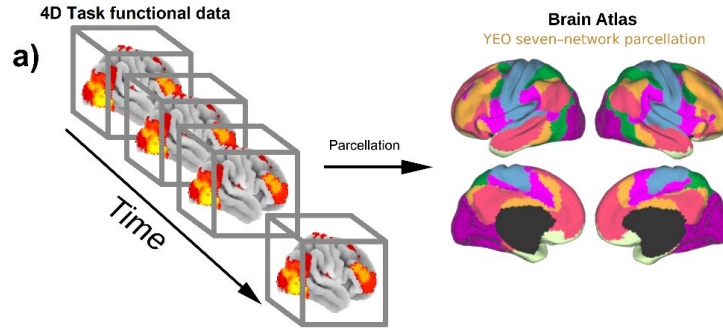
	precision	recall	f1-score	support
0	0.71	0.62	0.67	8
1	0.67	0.75	0.71	8
accuracy			0.69	16
macro avg	0.69	0.69	0.69	16
weighted avg	0.69	0.69	0.69	16

Feature Index	Feature Name	Importance Score
1676	Intraparietal sulcus LH - Inferior frontal sulcus	6.642316
238	Cingulate cortex posterior - Superior occipita...	5.738047
240	Cingulate cortex posterior - Middle frontal gy...	5.305342
618	Transverse sinus - Paracentral lobule superior	5.036077
415	Superior temporal sulcus with angular gyrus - ...	5.003628
235	Cingulate cortex posterior - Parieto-occipital...	4.847928
429	Planum temporale - Superior frontal gyrus medial	4.824347
102	Fusiform gyrus - Postcentral gyrus inferior	4.462734
1711	Dorsomedial prefrontal cortex antero-superior ...	4.408093
1229	Descending occipital gyrus - Superior parietal...	4.318116

Discussion

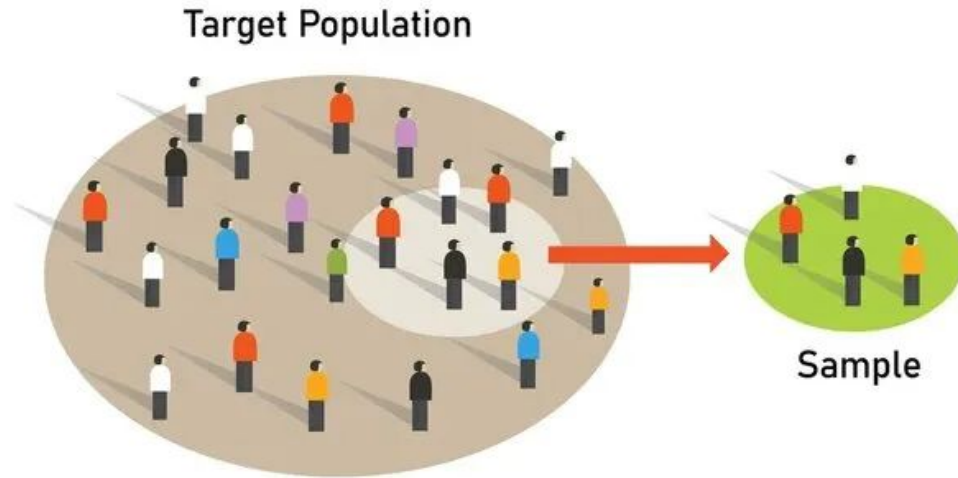
Potential: Machine learning shows potential for classifying ADHD using rs-fMRI data.

Model Performance: Higher recall for ADHD indicates distinct connectivity patterns in ADHD.

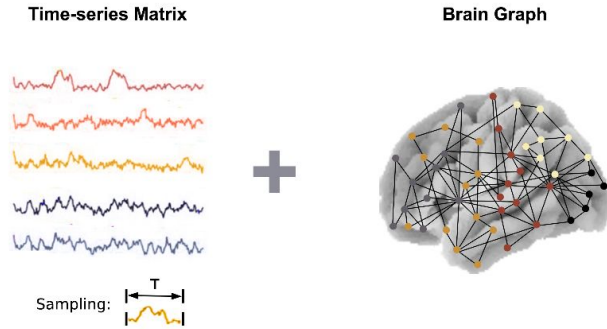


Limitations

Sample Size: Small participant pool limits generalizability.



Conclusions



Promise: Machine learning can potentially offer more objective and accurate diagnostic tools for ADHD.

Future Applications: Similar approaches could be applied to other psychopathologies to develop objective diagnostic tools.

Thanks for the attention!

